What's new in HTTP Live Streaming

WWDC 2026

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HLS Specification

The most recent specification of HTTP Live Streaming is RFC 8216. This document was published in August 2017. A more up-to-date reference is the current Internet-Draft draft-pantos-hls-rfc8216bis-22 "HTTP Live Streaming 2nd Edition".

For a preliminary version of the next update to the Internet Draft, please see (https://developer.apple.com/streaming/HLS-draft-pantos.pdf) posted on the developer website.

There is a related Internet-Draft for Content Steering, please see [Content-Steering] (https://datatracker.ietf.org/doc/html/draft-pantos-content-steering-05) for details.

The URL https://developer.apple.com/streaming/ contains links to the above documents.

EXT-X-PRELOAD-HINT for decryption keys

The EXT-X-PRELOAD-HINT tag now supports TYPE=KEY to hint at an upcoming decryption key. During live streams with high concurrent viewership, a key rotation can trigger a thundering herd of simultaneous key requests. By hinting at an upcoming key, clients preload it at a randomly-selected point between the time the hint is first seen and the key's first use, spreading requests across that window to reduce peak key server load and avoid playback stalls. For more information about preloading keys, see section [4.4.5.3 EXT-X-PRELOAD-HINT] (https://developer.apple.com/streaming/HLS-draft-pantos.pdf) of the preliminary version of the HLS specification.

Date Range Schedules

The Date Range Schedule is a mechanism that allows content producers to add multiple Date Ranges to an asset in response to a single EXT-X-DATERANGE request. It provides more flexibility for server-driven interstitial insertion.

A Date Range Schedule is an EXT-X-DATERANGE tag with a CLASS attribute whose value is "com.apple.hls.daterange-schedule". It has an X-URI attribute which references a JSON file that contains a list of Date Ranges. For more information about Date Range Schedules, see [appendix H] (https://developer.apple.com/streaming/HLS-draft-pantos.pdf) of the preliminary version of the HLS specification.

HLS Tools

Apple distributes several tools for segmenting and validating HLS streams. These are available from the Apple developer downloads site. A link is available at the URL above.

New platform support

HLS tools now support two additional Linux flavors in addition to Red Hat Enterprise Linux. All tools provide full feature parity across the following platforms:

- macOS
- Red Hat Enterprise Linux (version 9.5)
- Ubuntu (version 24.04.3 LTS)
- Debian (version 13.2)

AVMetrics support for AirPlay Video sessions

AVMetrics is now supported during AirPlay Video playback on Apple TV 4K and AirPlay-enabled smart TVs, requiring no changes to existing application code. The same AVMetricEvent classes available in HLS are fully supported during AirPlay Video playback. This delivers end-to-end observability of playback metrics across the Apple ecosystem in a unified manner, regardless of whether HLS content is played back locally or streamed via AirPlay Video.

Please refer to the [AVMetrics APIs]

(https://developer.apple.com/documentation/avfoundation/avmetrics) for more information.

AVPlayerItemSampleBufferOutput

You can use the new AVPlayerItemSampleBufferOutput API to retrieve decoded audio data during playback. This API can be used to scan ahead of playback so that your app can perform analysis to enhance the playback and/or navigation experience.

Please refer to the [AVPISBO APIs] (https://developer.apple.com/documentation/avfoundation/avplayeritemsamplebbufferoutput) for more information.

Live Activity for offline HLS downloads

Offline downloads through AVAssetDownloadTask now automatically display progress as a Live Activity while the initiating app is running in the background. This keeps users informed of download progress without needing to reopen the app. Multiple downloads from the same app are aggregated into a single Live Activity. A stop button on the Live Activity enables users to cancel queued and in-progress downloads without returning to the app.

Please refer to the [AVAssetDownloadTask APIs] (https://developer.apple.com/documentation/avfoundation/avassetdownloadtask) for more information.

HDR update

Dolby Vision 8.1 was supported in tvOS and starting with this year's release, macOS, iOS, iPadOS, and visionOS support Dolby Vision 8.1.

visionOS 27.0 and later will also support HDR10/10+ for MV HEVC (3D) and the Apple Projected Media Profile (APMP)

Auto generated Media Selection options

The CHARACTERISTICS attribute of the #EXT-X-MEDIA tag may include "public.machine-generated" to indicate the Rendition was authored or translated programmatically.

In all such cases, the localized display name of the corresponding AVMediaSelectionOption as shown in the standard UI for media selection that's provided by AVKit will be sufficient for end users to recognize their provenance.

For example, the display name in English for auto generated subtitle rendition with the LANGUAGE attribute "en" and a CHARACTERISTICS attribute that includes "public.machine-generated" but does not contain “public.translation”will be

English Generated ✨

The display name for auto translated subtitle rendition with the LANGUAGE attribute "es" and a CHARACTERISTICS attribute that includes "public.machine-generated" and “public.translation” will be

Spanish Translated ✨

Please note that for non AVKit based applications, ✨will not be part of the display name. Please see [Discover generated subtitles and subtitle styles] (https://developer.apple.com/videos/wwdc2026) for more information.

MTAudioProcessingTap for HLS

Instances of MTAudioProcessingTap can now process a mix of audio output from any AVPlayerItem, regardless of the number of audio tracks that produce audio during playback and regardless of their encoding formats, with the exception of FairPlay-protected audio.

To set up MTAudioProcessingTap, create AVAudioMixInputParameters with the special track ID, AVAudioMixInputParametersTrackMixID, and the instance of MTAudioProcessingTap as the value of the property audioTapProcessor. After setting up the special track ID and the property, create an AVAudioMix that includes your AVAudioMixInputParameters and set the AVAudioMix as the value of the AVPlayerItem property audioMix.

Unless a preferred processing format is specified, instances of MTAudioProcessingTap operate in a format that's optimized for delivery to the audio output device. You can specify a preferred format for a tap via use of MTAudioProcessingTapCreateWithPreferredFormat.

Developer portal update

New HLS sample for AV1 and an Interstitial is now available on examples page on the portal.

Archives

What’s new in HLS (2025) (https://developer.apple.com/streaming/Whats-new-HLS-2025.pdf)

What’s new in HLS (2024) (https://developer.apple.com/streaming/Whats-new-HLS-2024.pdf)

What’s new in HLS (2023) (https://developer.apple.com/streaming/Whats-new-HLS-2023.pdf)